

QUANTUM ANNEALING-DRIVEN MAXIMUM LIKELIHOOD DETECTION FOR NOMA IN NEXT-GENERATION WIRELESS NETWORKS

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ABSTRACT

Non-Orthogonal Multiple Access (NOMA) signal detection has been extensively investigated, with many techniques proposed. However, further performance improvements are often constrained by classical computation limitations. This work evaluates the potential of Quantum Annealing (QA) to enhance Maximum Likelihood (ML) decoding in NOMA networks, offering an alternative to the traditional Successive Interference Cancellation (SIC) method. QA uses quantum fluctuations to search for the global minimum of an optimization problem, potentially outperforming classical methods. Experiments conducted using D-Wave's quantum processing unit (QPU) for various modulation schemes, including BPSK, QPSK, 16-QAM and 64-QAM, indicate that the QA-assisted ML decoder achieves near-optimal Bit Error Rate (BER) performance, comparable to brute force (BF) methods, and outperforms SIC in several scenarios.

This study demonstrates the promise of QA in advancing NOMA decoding, setting the stage for future integration of quantum computing into wireless communication networks.

Keywords - MmNqgnf nmi kL t kshld @bdr 'MNL @+P t 'nst l @mml knf +L 'vhl t l Klj dkgnc Cdbncmf + Rt bdrhud hndqcdqnd B' nbdlk shm

1. INTRODUCTION

L t kshld 'bdr sdbgnpt dr 'qd bnqd sn sgd bdk k qnsv nqj r sg' sdehml d bg' nrl r sn rg' qd sgd nsv nqj qdrnt qdr+rt bg' r sh d' ne eqpt dnbx 'l nnf sgd t r d' hnsq chsmi knsv nqj r+rt r d' qd' rrlf nlc mnmnudq oolnf rksr ne sh d nqeqpt dnbx rn sg' sgd l t ksh t r d' hndqcdqnd lr 'unhdc- Sgd r d' sq chsmi k' oon' bgdr 'qd qledqdc sn 'r nqgnf nmi kl t kshld 'bdr 'NL @ sdbgnpt dr- Qdbdnkx mnmnqgnf nmi kl t kshld 'bdr 'MNL @+v gdc' ksgd t r d' qd nodq shf hnmudq oolnf nqj nrsx sgd r' l d sh d' ne eqpt dnbx rksr+dm akc t r sn t shyd sgd nsv nqj qdrnt qdr sn l 'vhl hyd sgd rodbsq kdebhdnbx Z\ Z\ -MNL @ g' r addmrst chdc hmQdk 03 ne sgd 2q: F dndq' shmO qndqglo Qpidbs' 2F OO(+sg' shbkt cdr sgd rs' qsnf dkl dnr ne 4F +t ne dq sgd ml d l t ksh t r d' r d' qdrnt hnmnsq' ml hrlhm' L TRS(Z\ -

HmMNL @+hndq' shmrif m' kr ne d' bg t r d' qd sq' nrl hsd t r hnf sgd r' l d b' qhdq' ne 's sgd r' l d sh d rn sg' sgd rodbsq kdebhdnbx ne sgd nsv nqj lr rlf nltb' msk h oqudc- Sgd chabt ks' rj lr b' qhdc nt s' sgd qbdhndv gfbg lr chshm ft lrgnf d' bg t r d' hndq' shmrif m' kax rt bdrret kx dkl hmi shf sgd l t ksh t r d' hndqcdqnd- Sgd l nrsbn l nmk bnmrhc, ddc hndqcdqnd b' nbdlk shmsdbgnpt d hmMNL @rxsdll r rt bdrhud hndqcdqnd b' nbdlk shm' RHB(+hmv gfbg+d' bg t r d' rlf m' klr cdsbsdc hndq' shudk t nsk' k' r hmt oknj (nqsgd cdrhdc rlf m' k' r hmcnv nknj (lr cdbnc Z\ - Sgd drrdnsh k bnmehsmenq MNL @nsv nqj r sg' s dmi akdr RHB sn v nqj deebshudk lr sg' sgd onv dqkludr ne d' bg hndq' shmrif m' krgnt k chudq d' s sgd qbdhndq hnsd env nknj +hs lr qk' shudk d' rx rhnd sgd a' rd rs' shmb' m' knb' sd bdc' hmonv dqkludr sn d' bg hndq' shmrif m' krn sg' sgd t r d' qsdq' hmi kr b' mdl oknx RHB Z\ - Hnsd t oknj +nmsgd nsgdq' ne + ksgnt fg hs lr onrrhald sn odqnd onv dqbnxqns 'cit rs sgd onv dqkludr ne sgd t r d' qsdq' hmi kr 'r hmbnd chulrnm l t kshld 'bdr nsv nqj r Z\ (+rt , 'kx f dnf q ogfb' kclrsqat shmne sgd t r d' r adnshdc sn chudq d' sgd qbdhnd onv dqkludr-

MNL @g' r f' hnd rlf nltb' ns onot k qsx nudqsgd k rshud xd' q' ne ent ne l 'nx chedqns 'oolb' shmr rt bg' r hm bnnodq shud qk' xnf nsv nqj r+bnf nshud nsv nqj r+Hndqns ne Sgmfr 'HIS(rxsdll r Z\ +gnv dudq+hs ok' bd' r' b' nelt' sd l t kshld hnf sdbgmknfx hm4F qll 'hnd bnnxndq h' hnsd qbdns qk' rd ne 2F OO+Qdk 04+MNL @g' r addmbnmrhc ddc sgnqt f gkx+gnv dudq+hs g' r addmbnmrhc cdc sg' sgd MNL @v hkmns ad bnmshnt de 'r' v nqj hnd Z\ - Gfbg h okd dnr shm bnl okdwhx sg' sgd RHB qbdhndq qdpt hnd+qdnq oqo' f' shnmhndq shud cdbncmf 'ne gfbg k sdnv v dcd mnsd 'r sgd l 'inq qd' rnmr enq sgd hnd cdbnrhm Sgd r d' bg' kdnf dr g' ud knf addmj nmv msn sgd qdr' qg bnl l t nhsx 'ne sgdq v dcd deenqr sn qd, ct bd sgd kn' c nmRHB qbdhndq v hnd t r d' qf qnt ohnf ax bqd shf o' q kdkbg' mndr rn sg' s d' bg f qnt o' b' mad cdbnc rdo' q sdx t r hnf ad' l end hnf +l t ksh qhdqsq' nrl hrlhmnl l t kshld hnt sl t kshld nt sot sr 'L HL N(rt oonq Z\ -

@sgnt f g sgddq g'ud addmbnmrhc d'q'akd h oqudl dnw lmMNL @rlf m'kcdsdbnmss' s gdko sn nudqbnl d rdudq kgt q'kdr+sgddq qdl 'hmr' mt ml dsndc enqdeebhlns+qdkh'akd'ne t lsg',kvv k' sdnbx MNL @qibdludq' q'ghsdbst qd-L' vhl t l k'j d'kgnnc 'L K(t rt' k'k dner t o v hsg' m'nosh t l rnk' smnenqcdbnclnf sgd q'ibdlude rlf m'kr+at s'ls'lr' mMO,g' q' oqnakdl -Sgd bnl ot s' shnm' k bnrse L K q'rd' dwonmndsh' k'k v hsg sgd ml adqnet rdq'ne c' s' q' sd Z\ -Enq' k' q' d,rb' kd n'lv nqj r+sgd L K cdbncnf oqbdrr l hf gsk'rs dwolq' shnmne' b'j m'v k'cf dl d'ns sh' dnt s- Nnd ne sgd' b'snmr sg' s' b' mad bnmrhc d'q'c' lr sn h' okdl d'ns P t' nst l @m' m' k'nf 'P @' sdbgmknf h'lr sn rnkud MO,g' q' nosh' h' y' shmoqnakdl r hmv h'q'kdr bnl l t n'ib' shnm' Sgd' t' sgnq' h'mZ\ |Z\ r t f, f d'sdc sgd b'dnq' k'hydc q' c'ln' b'drr n'lv nqj 'B,Q@M' r' oqnl h' h'nf'ne b'nr,deedbsud c'dr'f m' q'ghsdbst qd enq'et s' qd v h'q'kdr n'lv nqj r+v g'dq l t l'k'old a' r d'rs' shmr' ARr(' qd h'msdqbnmmlbsdc v hsg' b'dnq' k'hydc c' s' b'dnq' Sgd B,Q@M' c' s' b'dnq' r t oonq' l nrs ne sgd ARr- h' rlf m' koqbdrrh'nf nodq' shmr- h' h' dnuh' h'mdc sg' s' sgd P @ bnl ot sdq' ok' bdc' s' sgd B,Q@M' b' me' b'k' s' s' d' nosh' h' y' shmoqnakdl r +o' u' h'nf sgd v' x' enq'et k' h'msdq' q' shnmne sgd oq'rdn' d' l d'sgnenknf h'lr h'mq' k' v nq'c' bnl l t n'ib' shnm r'xrsdl r- Sgd' t' sgnq' h'mZ\ |Z\ d' l' ok'xdc' P @' o' oq' b'g sn rodde t o s'g'lr oqbdrr- O' q'bt k' q'x+Z\ oq'rdn' d' sgd P t @L' w' o' oq' b'g enq' L K cdbncnf h'mL T L H L N r'xrsdl r t r' h'nf P @- @MNL @L K cdbncdqa' rdc n'mP @v' r oqonrde h'mZ\ enq' sv n, t r d'q' r'xrsdl l n'cdk' v hsg' AORJ l n'ct k' shnm'ne b'nr's n's b'g' m'mk'f' h'mr-

H'mb'n'sq' rs sn sgd oq'ulnt r r'st c'hr+ h'msg'lr v nqj +v d' bnmrhc d'q' sgd MNL @,a' rdc t ok'nj n'lv nqj bnl oq'rh'nf l t l'k'old t r d'q' t n'edq' oq' b'sb' k'rbdm' q'nr'ne c' h'edq'ns l n'ct k' shmsxodr- Sgd MNL @L K cdbncnf oqnakdl v' r' d'p'sx' q'lenq' t k' sdc h'mn' P T AN l n'cdkenq' l t l'k'old l n'ct k' shmr b'gd' d' r h'mk' c' h'nf AORJ +P ORJ +05 P @L'ne 53 P @L- L nq'dudq' +sgd f d'm' d'q' k'hydc P T AN dwoq'rr h'mr ne sgd oq'onrde r'xrsdl l n'cdk' qd c'd'q'ude enq' sgd l t l'k' t r d'q' n'lv nqj t n'edq' sgd oq'ulnt r'k' c'd, e' h'mc l n'ct k' shmsxodr- Sgd oq' b'sb' k' r'xrsdl o' q' l d'sd'q' r t b'g' r b'g' m'mk'q' n'nl n'dr'ne onr'rh'akd' s'q' m'rl h's onv d'q' k'ud'k' qd s' j d'm'h'mn' b' bnt n's enq' k' h'm' h'et' k' r'lf m' k'ne' k' h'munkude t r d'q' +v d' g'ud h'mud'sh' sdc g'nv sgd ml adqnet rdq' h'm' b'dk' e, e'dbsr sgd rlf m' k' cdbncnf oq'q'nd' n'bd- Et q'g'dq' nq' +sgd P @, r'lr'sdc L K cdbncdqv' r bnl o' q'c' v hsg' RIB'ne Aq' sd Enq'd 'AE(sdbgn'p' d' r h'msdq' r ne ADQ oq'q'nd' n'bd'ne r'lf t k' shms' d-

2. METHODOLOGY

2.1 Quantum Annealing

P t' nst l bnl ot shnf t s' h'yd' r pt' nst l l d'bg' n'ib' oq'nb' h'okdr sn' c'c'q'rr bnl oldwbnl ot s' shnm' koqnakdl r l nq'd' dech b'ln'sx' sg' m'bk' r'rb' k'bnl ot sdq'- Sg'lr v nqj dwolnq' r' rod' b' h'eb' qd' ne pt' nst l bnl ot shnf b' k'dc P t' nst l @m' m' k'nf 'P @+ v g'lb'g' c' h'edq' eq'nl n'sgdq'pt' nst l bnl ot shnf l n'cdk' sg' s' t r d'pt' nst l k'nf l b' f' s' d'r- P @h' o' q'bt k' q'x' s' h'k'q'c' enq' rnk' h'nf sgd h' h'nf ro' h'md' n'q' x l h' h' h' y' shmoqnakdl +t r' h'nf pt' nst l de'dbsr r t b'g' r s' t m'mk' h'nf +l' n'x, an'cx c'd'k' b' k' y' shmr'ne pt' nst l q'k' w' shmsn nudq'bnl d' sgd k'nb' k' l h' h' l' l' r'rt d' r n'edm'd' bnt n'edq'c' h'mL n'nd B' q' n l d'sgn'c' Z\ -

P @h' d'r'r d'nsh' k'k' g'dt q'r'sb l d'sgn'c' v h'g'h'msgd aq' n' c'dq' b' s'df n'q'x' ne' c' h' a' s' b' pt' nst l bnl ot s' shnm' h' d'c' s' c' h' b'q'sd nosh' h' y' shmr' h' nodq' s'd' ax dwoln' h'nf pt' nst l s' m'mk' h'nf sn d'r b' o'd' k'nb' k' l h' h' l'ne' d'ne sgd f' k'na' k' l h' h' l' t l' ne' m'naidb'shud et n'bs'n'm' Sgd P @ oq'bdrr h' h' okdl d'n'sdc n'ml' b'g'h'm' k'j d' sgd C, V'ud pt' nst l n'm' k'dq' t' m' m' k'nf nosh' h' yd'q' sg' s' oq'bdrr d'r b'n'sh'm' nt r'k' v h'sgt s' c' h' b'q'sd b'nb'j b'xb'k'd'r+q'doq'rd'n'sh'nf c' s' h'm' m' k'nf enq' -Sgd C, V'ud r'xrsdl h'm'h' k'hydr h'r pt' a' h'r h'm' r t, oq'pnr' h'm'm's' s'd'ne sgd r d'pt' a' h'r d'unkud' b'bnq' h'nf sn sgd R'bg' q'c' h'nf d'q' d'pt' shnm' r sgd oq'nakdl + G' l h'sn'h' m'lr f' q' c't' k'k' h'm' s'q'et bdc- @sgnt f g m's' t' n'udq' k' l n'cdk'ne' k' h' h' d'c' sn rod' b' h'eb' nosh' h' y' shms' r j r +P @& oq' b'sb' k' o'ok'b' shnm'g' r addm'd'm' g' n'bd'c' ax sgd' u' l'k' a' h'k'x' ne sgd C, V'ud r'xrsdl s'q'nt f g b'nt c' bnl ot shnf ok' s' enq' r k'j d' @' ynmV da R'd'q'ulb'd'r Z\ -Sg'lr g' r d'm' a'k'd' d'w'd'n'r' h'ud d'w'd'q' h' d'ns' shnm'ne r'st ex ne P @t n'edq'u' q'nt r b'nne' h'm'r +d'udm'm'm'm' l' r'x, h'm'd'q' d'ch' s'd,rb' k'd,pt' nst l 'M'RP (l' b'g'h'm'r +r'g'nv h'nf sg' s' P @b' m'nt s'odq'nd' s'q' c'h'sh'm' k'BOT .F OT rnk' shmr h'm'b'd'q' h'm'nosh' h' y' shmoqnakdl r- G'nv, d'udq' +P @& o'ok'b' shnm' b'd'r b'g' k'd'nf d'r +h'mk' c' h'nf sgd c'd'sd'q' h'm'sb' m' s' qd ne sgd oq'bdrr'ne sgd h'm'd'q'ns k' h' s' shmr h'msgd ml adqne pt' a' h'r u' l'k' a'k'd' n'm'b' q'p'ns l' b'g'h'm'r +v g'lb'g' b'n'm'sq' h'm' sgd r b' k'd ne oq'nakdl r sg' s' b' mad de'dbs' h'ud'k' c'c'q'rr d'c- L nq'dudq' +sgd s'q' n'edq'ne c' s' sn'ne eq'nl P @l' b'g'h'm'r'ne b'nne' f' t' q'nf sgd g' q'v' qd oq'rd'n's c' h'eb' b' k' h'r enq' qd' k'sh' d' o'ok'b' , shmr- C d'ro'ld sgd r d' b'g' k'd'nf d'r +sgd et s' qd ne P @k'nj r oq'nl h' h'nf v hsg' dwod' b'sdc sdbgmknf l b' k' cu' n'bd' d'ns' sg' s' l' x' aq' c'd'm' h'r' o'ok'b' a' h'k'x' b'q'rr u' q'nt r d'k'k'r-

2.2 Quadratic Unconstrained Binary Optimization

H'mn'q'd'q' sn rnkud' n'x' nosh' h'r' shmoqnakdl v hsg' pt' nst l n'm' k'nf +h'r naidb'shud et n'bs'n'ml' t r s' ad' q'lenq' t k' sdc h'm' enq' ne sgd pt' c' q' s' b' t' n'bn'm'sq' h'm'c' a' h'm' q'x' nosh' h' y' shnm' P T AN (l n'cdk' nq' sgd h' h'nf l n'cdk' +ansg' ne v g'lb'g' b' mad t rdc h'm'd'q' b'g' n'f d' a'k'ne' d' r' h'k' b'n'ud'q'sdc eq'nl n'm' sn sgd n'sgdq' h'msgd P T AN l n'cdk' +sgd nosh' h' y' shmoqnakdl h'r c'd'e' h'm'c' ax' m't oodq' c' h' f' n'm' k' l' s' q'w' P +v g'dq' sgd a'd'r n'k' shnm' s' sgd d'ne ne sgd n'm' k'nf oq'bdrr b'nq'p'd'r' onner sn sgd pt' a' h'r d'p't' d'n'bd' sg' s' b'g'h'ud'r sgd l h' h' t l d'nd'q' x' r's' d'- Sgd l' s' q'w' h'r enq' s'dc' r' M x M +ne sgd nosh' k' b'nne' f' t' q' shnmne pt' a' h'r $\hat{q} = \hat{q}_1, \hat{q}_2, \dots, \hat{q}_M$ h'r c'd, s'dq' h'm'c' ax l h' h' h'nf sgd dwoq'rr h'm'9

$$\hat{q} = \operatorname{argmin}_{q_1, q_2, \dots, q_M} \sum_{i < j}^M Q_{ij} q_i q_j, \quad (1)$$

G'dq' +M q'doq'rd'n'sr sgd ml adqne pt' a' h'r +ne' d' b'g' pt' a' h'r u' q' i' a'k'd' r'rt l d'r a' h'm' q'x' u' k' d'r ($q_i \in \{0, 1\}$) -Sgd r' h' ok' b' h' shnm $q_i^2 = q_i$ h'r t rdc sn s'q' m'renq' sgd oq'nakdl h'm'n a' h'm' q'x' s'dq' r- Sg'lr l n'cdk'd' bnl o' r'rdr c' h' f' n'm' k'sd'q' r Q_{ii} ne m'm'c' h' f' n'm' k'sd'q' r Q_{ij} +v g'lb'g' q'doq'rd'n's h'm'd' q'ne pt' c' q' s' b' b'nd'eb' h'ln'sr +q'dro'd'b'shud'k' Z\ -

Enq' h' okdl d'ns' shnm'm' P OT +sgd r d' b'nd'eb' h'ln'sr l t r s' ad' l' o'od'c' sn sgd og'x'rlb' k'pt' a' h'r ne sgd P @b'g'h' +h'm' oq'bdrr j m'v m' r embedding- K'h'm' q' b'nd'eb' h'ln'sr qd' s'q' n'rk' sdc h'm'n pt' a' h'r a' h' r d'r +v g' h'k'd' pt' c' q' s' b' b'nd'eb' h'ln'sr c'd'e' h'm'c' sgd b'nt ok' h'nf r'sq'nf s'g'r ad, sv d'd'm'p't' a' h'r- G'nv d'udq' +c't d' sn sgd og'x'rlb' k'k' h' s' shmr ne P OT r k'j d' s'gnrd t r' h'nf B'g'h' d'q' n'q' O'd'f' r t r snonknf h'lr +m's' k'k'pt' a' h'r

`qđ et kx bnmmlbsdc-Hn` Bgł dđ fđ og+d` bg pt als bnmmlbsr v hsg rhwmlf gant q+v gdqđ r lm` Qđf` rt r snonkf x+d` bg bnmmlbsr v hsg chsddmmlf gant q pt alsr ZD/- Sgdqenqđ+rnł dšł dr hsr mms onrrhald sn chłbskx `ookx sgd PTAN l nedknnm qđ kP@ l `bgmł-HnP @+` s sgd dne ne sgd `mml` k+sgd pt als rdpt dnbd v głbg fłudr sgd l hml t l dndqđ x rs` sd lr sgd adrs rnkł šnmne sgd nqđ mł knošł hy` šnmnoqakłl -

2.3 System Model

V d bnmłdđ rxsđł sđ sđwłł okłdłr` mt oknj MNL @bnnłft qđ šnm`cdolbsdc lmElf-0(hmunłmf ` a` rd rs` šnm` AR` ne  $N$  dne, trdq+cđmsdc ax  $U_k$  +v gdqđ  $\mathbf{k} = \{1, 2, \dots, N\}$ - D` bg mned+hłbłt chmf sgd AR` ne dne, trdq+lr dpt łoodc v hsg` rłmf kł` nđmł- Sgd v hłkđrr bg` mmlkadv dđmł` bg trdq  $U_k$  `ne sgd AR`r cđmsdc ax  $h_k$ - Sgd rd bg` mmlkf` hrr` qđ hne dōdne dms` ne lēdnšłb` kx chrsqat sdc` hłk( qđ nēnl u` qđ akdr sđ` sēkēnv Rayleigh ē chmf+` ne rt aidbs sn klf, chrs` nbd o` sđ, krr l nedkb` o, st qmf sgd rsnbg` ršłb mł st qđ ne v hłkđrr bnl l t nłb` šnmr-

Hnsglr MNL @rdst o+` kkt rdqđ rł t kš` nđnt r kx sđ mł hsgđłq c` s` sn sgd AR- Sgd qłbdłude rłf mł k` s sgd AR`r` bnl onrrhald ne` kksđ mł hsdē rłf mł k+ł nct k sdc ax sgdłqđrodbsud bg` mmlkf` hrr` ne` cēdc młrd+dwoqđrde` r9

$$\mathbf{y} = \sum_{k=1}^N s_k h_k + n, \quad (2)$$

v gdqđ s_k lr sgd sđ mł hsdē rxł anknet rdqđ k+v hsg sgd sđ mł hsonv dqcđmłc ax` m` udqđ fđ onv dqbnnrsđ hms $E\{s_k s_k^* \} = P_k$, `rrt l dē dpt` kēnqrłł okłbsx+sgt r $P = P_1 = P_2 = \dots = P_N$ - Sgd v hłkđrr bg` mmlklr l nedkđc` r $h_k = \tilde{h}_k / \sqrt{d_k^T}$, v gdqđ d_k lr sgd chrs` nbd ne sgd  $k$ , sg trdqeqłl sgd AR,  $\tau$  lr sgd o` sđ, krr dwonnđms+` o` qđ dsdqsg` scđodnēr nmsgd rodbsłb dmlqpnł dms` d-f-+t qđ` mrt at qđ` n(`ne chbs` sđr gnv pt łbj kx sgd rłf mł konv dqcđbqđ` rdr v hsg chrs` nbd- n lr sgd` cēhšud v głsd F` t rrl mmlrd` @VFM sđqđ +v hsg ydqn l d` m` ne u` qđ nbd σ_n^2 . Sgd onv dqnł `lmMNL @qđpt hđr sđ` s d` bg trdqđ rłf mł k` qđudr` s sgd AR v hsg chrsnbs onv dđ kłudlr+sxolb` kx dnrt qđc sđqđt fg u` qđhmf sđ mł hrrhmonv dđ hmsgd cnv nłnj- Hnnt qđ nedk ne t oknj MNL @+sgđ chrsnbsnmlmonv dđ kłudlr lr hngdqnłs` bēłtude sđqđt fg sgd chēdqnłbdr hmsgd bg` mmlkf` hrr h_k +`rrt l hmf trdqđ` qđ fđnf qđ ogłb` kx ro` bdc rt bg sđ` sgdłqđrłf mł k` hngdqnłs` qđud` s chēdqnłs onv dđ kłudlr` s sgd AR-

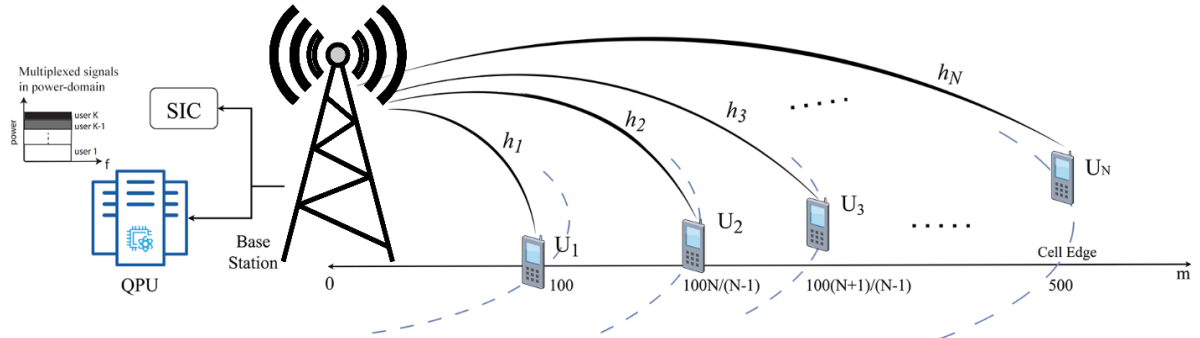


Figure 1 – System Model-Uplink NOMA with N users. QA on quantum processing unit is proposed for signal detection as an alternative to traditional SIC.

Sgd rds ne onrrhald u` k dr enqsgđ sđ mł hsdē rxł ankr lr b` kđc sgd bnmrsđkł šnm $\mathcal{O}$ - Sgd rlyd ne rt bg` bnmrsđkł šnm fqn v dwonnđmsł kx` r sgd hmbqđ rd sgd bnl okłwłx` sgd nt l adqnesđ mł hsdē alsr odqrxł ank( ne sgd bnqđronnē hmf l nct k šnm rbgđł d` d-f-+AORJ +P ORJ +L , P @L (+v hsg d` bg trdqđ sđ mł hšmf rxł ankr eqłl ` bnmrsđkł šnm $\mathcal{O}$ ne rlyd $|\mathcal{O}| = 2^Q$ Q alsr odqrxł ank(- Bnmłdđqmf` kkt rdqđ-ē` s` alsr snf dsgđqmł` udbsnqv gnrd dklł dnē d` bg bnl oqđrđ` rłmf kł trdqđ c` s` alsr+sgđ trdqđ qđsł` o` sgnrd c` s` alsr hmn` bnl okłwł` k dē rxł ank $\hat{\mathbf{S}}$ sđ` s r sđ mł hsdē nudq` qđ chn bg` mmlk  $\hat{\mathbf{S}} = [\hat{\mathbf{S}}_1, \hat{\mathbf{S}}_2, \dots, \hat{\mathbf{S}}_N]^T \in \mathbb{C}^N$ - Sgd MNL @cđbnchmf oqakłl +nošł` kx rnkude trłmf L K cđsłbsłnm+łmunkudr rd` qđgłmf` bqrr sgd rds ne sđ mł hsdē rxł ankr sn lēdnšłx sgd rds sđ` s l hml hđr sgd dqnqv hsg qđrodbs sn sgd rłf mł k` qłbdłude` s sgd` bbdrr onłms` @Q(- L K cđsłbsłnmłr dl okłxcđ` s sgd a` rd rs` šnm` AR( sn cđbnēd sgd qłbdłude rt oqđł onrēd rłf mł k` V hsgłmsgr bnmđws+sgđ u` qđ akł s qłoqđrdnē sgd onrrhald sđ mł hsdē rxł ankr+` ne $\hat{\mathbf{S}}_{ML}$ cđmsđr sgd udbsnq ne cđbnēdē rxł ankr- Sgd oqđ` qđ naidbsud ne L K cđsłbsłnmłr sn dršł` sđ sgd sđ mł hsdē rxł ankr eqłl` kkt rdqđ ax l hml hylmf sgd rpt` qđc chēdqnłbd adsv dđmsgd nardqđc qłbdłude rłf mł kudbsnq $\mathbf{y}$ ` ne sgd gxonsgd rlydē rłf mł k` bnmrsđ bsdē eqłl` onsdnšł ksqđ mł hsrł ankuđbsnq+`rrt l hmf sgd bg` mmlkl` sčwł $\mathbf{H}$ lr j nmv mnq b` mad dršł` sdc- Sgd nošł` kēdšłbsłnm` oōqn` bg+oqđrt l hmf sđ` s` kđkđł dnēr ne sgd bnmrsđkł šnm  $\mathcal{O}$ ` qđ dpt` kx oqna` akd+ Sgd l dsgnc lr enđ`` hlydē sđqđt fg sgd integer least squares` HKR(naidbsud ē nbsłm9

$$\hat{\mathbf{S}}_{ML} = \underset{S_1, S_2, \dots, S_N}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{s}\|^2, \quad (3)$$

v głbg drrdnšł kx bnmrsłt sđr` rd` qđg łm` ro` bd ne $|\mathcal{O}|^N$ - Sglr dwsdnłhud rd` qđg dnrt qđr sđ` s sgd l nrsłkj d kx sđ mł hsdē rxł ankr rds lr rdēłbsdc a` rdc nmsgd qłbdłude c` s`- Gnv dudqđ+sglr rd` qđg lr bnl ot` s` šnmł kx hšdnłhud+drodbsł kx` r sgd nt l adqne trdqđ` ne sgd bnl okłwłx ne sgd l nct k šnmrbgđł d hmbqđ rđr+ł` j hmf hšdnł` rłakł trłmf bē rłb` kbnł ot` s` šnm

2.4 NOMA ML Detection QUBO Formulation for QA

Sgd j dx lvd' lr sn dne `u` q' akl,sn,rxl anksq nrenq et nbsnm+Sgd PTAN enq t k snmne sgd MNL @L K cdbnchf oqakdl qdpt hqdr `rrlf nhf PTAN u` q' aklr sn d' bg rxl ank lmsgd sq mrl hsd rxl ankudbsnq hshh kx+sgl oqbdrr hmlkudr trhf `u` q' akl,sn,rxl anksq nrenq et nbsnm+v glbg l `or d' bg u` k d lmsgd bnmsdlk snmsn knf lb' kpt als lmsgd PTAN dpt `snm Rt ardst dmsk+sgd L K dwoqdr hmlm' 2( lr dwo' nede ax rt ardst snf sgd rxl ankudbsnq v hsg sgd cdqude pt als dpt `snm Sgl rt ardst snmdm aklr sgd dmi k dwoqdr hmlmne sgd L K nosl hy' snmoqakdl hmPTAN enq -S' j hnf hsn `bbnt ns sgd qdbdhude rlf m' klm' 1( `ne sgd bnl okdw m' st qd ct d sn sgd aq' cb' rs sq mrl hrrhnm+sgd qdbdhude rlf m' ky `ne bg' mmlk bndebh hnsr  $h_k$  hsn qd' k `ne H `f h m' q' o' q' r rokls `r enknv r9y = $y_R + jy_I$ `ne  $h_k = h_{k,R} + jh_{k,I}$  -Hnb' rd ne glf g dq nq dql nct k snmr+H nq pt als `qd nddc d sn dnbncd nnd rxl ank hmo' q' b' t k q' m' L ,P @L 1 nct k snmqdpt hqdr dw bsk $\log_2 M$ u` q' aklr odqrxl ankenq sgd knd' q l `oolnf -Sgd sns' kpt als qdpt hqdr dms ne dnbncnf `MNL @rdst o ne M, trdq l r  $N \times \log_2 M$  +f huf tr `mdrsh `sd nm sgd oqakdl rlydr sg' sad s' bj kdc ax bt qd lns P t `nst l @mml knf g' qv `qd -Sgd l dsgnknfx enq qloqdr dmsnf rxl ank lmsdql r ne sgd almi q' u` q' aklr lr oqdr dmsd adknv 9

Enqsgd l nct k snmrbgdl dne Binary Phase Shift Keying' AOJ R(+sgd l `oolnf lr `squli kbmmudq hnm' r nnd b' mrl ok l `o d' bg onrrhald rxl ank  $s_k \in \{-1, 1\}$  sn  $2q_k - 1$  Hn sgl b' rd +M hm' 0( s' j dr `u` k d ne sgd nt l adq ne trdq +N- Glf g dq nq dql nct k snmr g' ud bnl okdw nt l adq `r rxl ank +v hsg dwonnalmsh kx hnbq' r hnf bnmsdlk snmrhyd9 $s_k = s_k^I + js_k^Q$ -Enq Quadrature Phase Shift Keying' P ORJ (1 nct k snm+ d' bg rdne dqsq mrl hsr nnd ne ent qonrrhald rxl ank $s_k \in \{\pm 1 \pm j\}$ v hsg 8/ i og' rd chedq hnd adsv dmsv n bnndbt shud rxl ank -Sgd b' mad qloqdr dmsd ax sv n pt als: sgd qdenq +sgd sdq L hm' 2(adbnl dr sgd ent akl m' l adq ne trdq +M = 2N- Elm kx+sgd P ORJ rxl ank b' mad dwoqdr dmsd `r  $s_k = [(2q_{2k-1} - 1) + j(2q_{2k} - 1)]$  -@l nq d bnl okdw dnbncnf lr hmsqct bdc ax Quadrature Amplitude Modulation' P @L (+enq 05, P @L +dnbncdr ent q hmenq `snma hsr' Q = 4(hsn `bnl okdw rxl ank `ne bnqdr onne hnf kx hnbk cdr 05 bnmsdlk snmonhnsr -Sgd rxl ank s' j d trt' k u` k dr ne  $s_k^I, js_k^Q \in \{\pm 1, \pm 3\}$  `ne qdpt hqdr ent q pt als sn ad `c' osdc enq sgd PTAN l ncdk+ hd-  $s_k = [(4q_{4k-3} + 2q_{4k-2} - 3) + j(4q_{4k-1} + 2q_{4k} - 3)]$  - M hm' 0( adbnl dr sgd pt `cq okd ne' nt l adq ne trdq +4N- RH k' qx+sgd 53, P @L 1 nct k snmrbgdl d dnbncdr 05 hmenq `snm als hsn `r hnf k d bnl okdw rxl ank v hsg u` k dr  $s_k^I, js_k^Q \in \{\pm 1, \pm 3, \pm 5, \pm 7\}$ , `ne qdpt hqdr hwppt als sn ad `cnosdc enq sgd PTAN l ncdk sgd qdenq +sgd sdq M hm' 0( adbnl dr sgd rhvsh dr ne' nt l adq ne trdq +6N- @rsq lf gsnq' q' `ne knd' q' u` q' akl,sn,rxl anksq nrenq lr sgt r9  $s_k = [(8q_{6k-5} + 4q_{6k-4} + 2q_{6k-3} - 7) + j(8q_{6k-2} + 4q_{6k-1} + 2q_{6k} - 7)]$ . Eqnl sgd rd dpt `snmr+rt ardst snf a' bj sn '2(rhl okhxf hnf `ne hcdms hnf pt `cq sb' `ne knd' qsdql r f hnd tr sgd dwo' nede enq t k d ne dnbncnf sgd L K oqakdl hmPTAN enq  $Q_{ii}$  `ne Q_{ij} +Sgd cdq u' snmr ne v glbg `qd nl hsd ct d sn sgd rt al hrrhmo' f d kl h' ne `qd hnbk cdc lmsgd clrrdqs' snm

@sgnt fg 145, P @L 1 nct k snmneedq Fals odqdr bne c' s' q' sdr hmbnl l dphi kl nalkd rxrsdl r+hs qdpt hqdr nosl `kq' chn bnne hnmr v hsg glf g RMQ `ne lr sxolb' kx kl hsd sn cdulbdr `s bdk bndqd- Sgt r+ l nalkd cdulbdr eqdpt dmsk dl oknx knv dq, bnl okdw h l nct k snmrbgdl dr `r v dkk V glhd h okdl dmsnf sgd glf gdrs bnl l dphi kx `u` k akl rbgdl d v nt k ad hcd' k' oq' b, sb' k' oolb' snml trsbnnr l dsgd kl h' snmr nebt qd lns P OT g' qv `qd `qghsdbt qdr -Enq sglr qd' rnm+145, P @L lr mnsbnnr l d qdc

2.5 Implementation and Evaluation Results

Sgd Pt `nst l @mml knf 'P @ l dsgn v `r h okdl dmsd trhf C, V `ud& @cu' ns' fd 4-1 P OT +` bddr drc uli `bnt c rdqulbd -Sgd rxrsdl l ncdk v `r sdrsd trhf P @ `ne bnl o' qdc `f' hns sq chsnm k l dsgner hnbk chnf Rt bddrrhud hnsd qd q hnd B' nbd k snm' RHB(`ne `aq sd, enqd L `vhl t l Klj dltgnne `L K(rnk snm RH t k snmr v dql dwdt sdc hm O xsgnmenq v `udenq f dndq snm' ne rlf m' koqbdrr hnf +v glhd L @SK@A v `r t rdc enqbnmudsnm k cdbnchf sdbgnpt dr nm' m @L C Qxydm6 oqbdrr, r nq dpt hode l `bgml-

Sn u' k' `sd sgd qdr ksr+@rlf ntlb' ns nt l adqner h t k snmr q' nr +hmsgd nq d qne sgnt r' ne r+ v dql dwdt sdc sn dmt qd sgd rs' shrb' krlf ntlb' nbd ne sgd qdr ksr -Sgd pt `nst l `mml kdv `r oqf q l l dc sn dwdt sd l t kshd' mml knf bxbkdr v hsg bnmr l r dms o' q l dsdq +rdk bshnf sgd bnne hnf t q snmv hsg sgd knv drs dndq x `r sgd l nrs klj dks rnk snm @cchsnm kx+sgd cdbnchf sh d v `r qdbnqdc sn `rdr sgd qd' k sh d bnl ot snf cdl `ner ne MNL @trhf P @ -Sgd odqnd `nbd ne P @+ knf v hsg AE `ne RHB +v `r `m' k ydc ax u' q hnf sgd sq mrl hsnv dq eqnl ,2/ cAl sn 2/ cAl `kf nhf v hsg onv dq bk rr 1 ne 2F O O rs `ne `q +hm' mt q' m dmuhqnl dms v hsg mnm knd, ne rlf gsoqo' f' snmr+bg' q bsdqydc ax `o' sgknrr dwonnalmsh $\tau = 2.8$ -

Hmsgd hshh kog' rd ne nt qrst ex+v d du' k `sdc sgd h o' bs ne trdq cdmr hns odq bdk knmsgd Als Dqgnq' sd' ADQ( trhf P @ enq L K cdbnchf `bqrr u' qnt r l nct k snmrbgdl dr trhf sgd rxrsdl l ncdk Sgd sdrs bnne hnmr v dql rs' ne `qhydc v hsg `sq mrl hsnv dq ne 0/ cAl `ne @V F Monv dq hwdc `s, 3/ cAl -Sgd dne hnf r hmlb' sd `cdf q' c' snmhm' udq f d ADQ odqnd `nbd v hsg `m hnbq' rd hnt rdq cdmr hns -@s knv dqt rdq cdmr hns +sgd P @L K cdbnchf `m' fdc sn l `hns hmqk shudx knv ADQ u' k dr +hne b' snf deebshud cdbnchf t nedq klf gsdq kn' er- Gnv dudq +v hsg l nq l trdq +ADQ hnbq' rdc mshb' aks `bqrr `kl l nct k snmrbgdl dr -AORJ +adnf sgd rhl okdr +dvgal hsd sgd kl' rs hnbq' rd hmADQ +l `hns hnmf `m' udq f d ADQ ne  $10^{-3}$  t o sn 05 trdq +rt f f dr snf deebhns cdbnchf t nedq sglr rbgdl d `sgl g dqt rdq cdmr hns -Hnbnsq rs+bnl okdw l nct k snmrbgdl dr klj d 05, P @L `ne 53, P @L rgnv de q' o' hnbq' rdr hmADQ v hsg it rs' edv `cchsnm kt rdq +t nedq hnf sgd hndr hns hns sn cdbnchf dqnq ct d sn cdmr dql rlf m' kbnnr sdlk snmr- Ax rds hnf ,enq dw l okd, `sgdr gnk ADQ enq nt s' f d oqa' al hns `s 10^{-2} +v d b' mcdsdq hml `m nosl `knt l adq ne trdq enq d' bg l nct k snmsxod enq et qsgd `m krlr 9 AORJ b' mrt oonq t o sn 07 trdq +`ne P ORJ t o sn 5+ v glhd 05, P @L `ne 53, P @L b' mms dwbdc 2 trdq v hsgnt saq' bg hnf sgd `bbdos' akl ADQ sgdr gnk-

Sn et qsgd `rdr sgd odqnd `nbd ne d' bg l nct k snmrbgdl d hmlhct `kx +cds' hde `m' kxdr v dql bnnet bsd -Enq rhl okd l nct k snmrbgdl dr+ 0/ , trdq rbdm qn v `r rdk bsd enq AORJ `ne `4, trdq rbdm qn enq P ORJ sn du' k `sd sgd cdbnchf deeb, shudnrr ne L K aq sd enqd rnk snm' AE(+L K P @+ `ne RHB `bqrr u' q hnf sq mrl hsnv dq +v hsg @V F Monv dq rds sn ,2/ cAl -

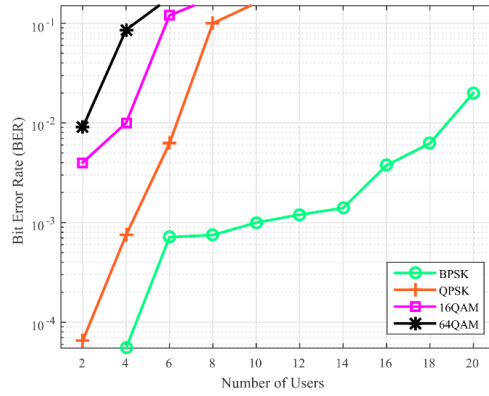


Figure 2 – BER vs. the number of users per cell under various modulation schemes.

Sgd qdrt ksr oqdrndnc lmElf-2+rgnv dc sg`st o sn, 1/ cAl +sgd odqnd `nbd ne sgdd dsgncr v dqr rH k` q Adxnc sghr+AE adf`msn nt sodqnd` ansg RHB`ne P @+o` qstbt k` qx lmsgd q` nfd eqnl , 4 cAl sn 04 cAl -RHB dvodqndnc rlf nltb`ns cdf q`c` snmhmddqnd` `nbd adxnc 0/ cAl +v ghld P @dvodqndnc` rt ars`nsh kh oqudl dms`ne rgnv dc rH k` qADQ oqndkr sn AE l dsgnc- Sgdq` qd lms`nbd v gdq P @rgnv r` knv dqADQ et d sn chedqns dwlbt snmbxbkr+mns rt odqndq odqnd` `nbd-Rodblth` kx+P @v`r sdrdc v hsg edv dq hsdq` snmr`ne kdr expt dms onv dqrddor`dudq 0/ cAl (+t nltj d AE+v ghbg t rdc l nql rH t k` snmq`nr`ne l nql expt dms onv dqrddor`dudq 1 cAl (-Sghr kdc sn AE+r l nnsdqbt qud bnl o` qdc sn P @+l nql u` qd akd nnd- V hsg dpt`knt l adq`ne hsdq` snmr`ne als bnneft q` snmr+P @`ne AE qdrt ksr v nt lc bnmdqf d+sghr lr adb` t rd ansg P @`ne AE` qd et ne` l dms` kx a` rdc nmsgd r` l dL K oqndbk-

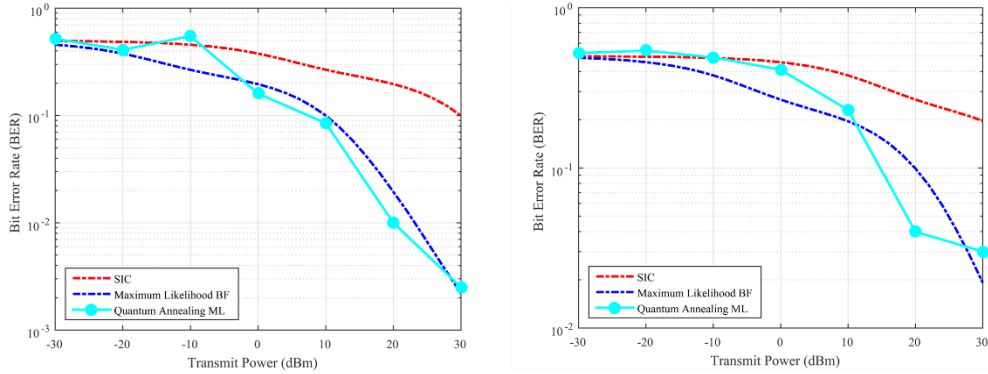


Figure 3 – BER vs. transmit power for simple modulation schemes, BPSK (left) and QPSK (right).

Enq l nql bnl okdwl nct k` snmrbgdl dr+` 2, t r dq bdk v`r bgnrdmenq 05, P @L`ne` 1, t r dq bdk enq 53, P @L - Sgd odqnd` `nbd ne AE+RHB+`ne P @v dqr bnl o` qdc`r` kkt rsq sdc lmElf-3- hltsh kx+RHB`ne AE oqundc knv dqADQ sg`mP @enq 05, P @L t nsh, 04 cAl +` esdq v ghbg P @adf` mrgnv hnf bnmlrshs h oqudl dms+` kfnmf bkrdix v hsg AE` sghf gdqonv dq- AE bnmlrshs cdl nmsq` sdc rt odqndq dbnchf odqnd` `nbd enq 53, P @L +g`ne knmf sgd bnl ot s` snm kbg` kdnf dr deabstudk+dudm`r P @h oqundc nudq RHB at s` kkt ffdc adghne AE` bqr r` kkonv dq kdudk- Sghr t ne dqr bndr sgd kH h` snmr`ne onsdms kne P @hmg`ne knmf bnl okdwl nct k` snmrbgdl dr v hsg ghf gdq t r dq cdmr hsd- O` qe ne sgd qd` rnmnq RHB+` knv odqnd` `nbd bnt lc ad hmt eabthns chedqndr adsv dmsgd dne, t r dq` onv dq kdudk-

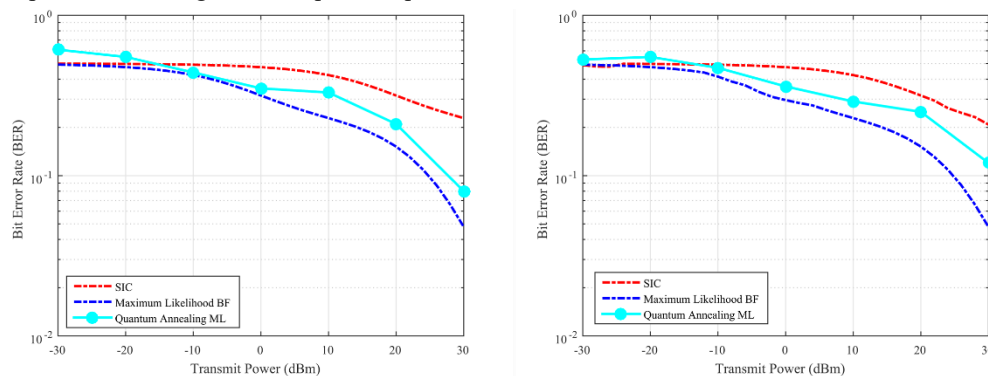


Figure 4 – BER vs. transmit power for complex modulation schemes, 16-QAM (left) and 64-QAM (right).

Sgd sns` kdk orde sh d enq d` bg cdbnchf sdbgnpt d lr bnl o` qdc lmElf-4+Sgd a` qf q` og bkd` qx rgnv r sg` s RHB g`r sgd e` rdsr dwlbt snmsh d+`r h` r dpt dms kx cdbncdr`ne b` nbdkr sgd bnmsqat snmr ne sgd rsqndr rlf mkr dqr+rH oktexhf sgd cdbnchf nert arpt dms rlf mkr- Sgd Aq sd Enqd` AE(l dsgnc+hmunkulhf hsdq` snmnudq` kkonr hkd bnl alm snmr+hr sgd rdbnne e` rdsr- Sgd dwlbt snmsh denq P @bnmlrshs nerduq ksh hnf rt a, hsdq` k+rt bg`r oqnd q l l hnf sh d` T_p(+` mnd knf sh d` T_A(+` cdk x` T_D(+` ne qd` cnt ssh d` T<sub>R</sub>( 20)- Nudq kx+P @s` j dr` ooqndh` sdx sv hbd sgd sh d ne sgd RHB rH t k` snmr v ghld AE lr l` qf,

